

HUMBERSIDE, United KingdomWMO: **033735**Lat: **53.574N**Lon: **0.351W**Elev: **37**StdP: **100.88**Time Zone: **0.00 (EUW)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.2	-5.1	2.5	-0.8	-3.8	2.8	0.2	14.7	10.9	13.7	9.7	2.8	210	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	26.0	18.8	24.1	17.9	22.8	17.1	19.7	24.0	18.7	22.7	17.9	21.5	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.5	17.1	12.3	20.3	16.2	11.6	19.3	56.2	24.1	53.1	22.8	50.5	21.7	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.7	11.3	10.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-6.0	29.7	4.2	1.8	-9.0	31.0	-11.4	32.1	-13.8	33.1	-16.8	34.4		
			-6.4	21.8	4.1	1.0	-9.3	22.5	-11.7	23.0	-14.0	23.6	-16.9	24.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	4.6	5.1	7.1	9.3	11.9	14.7	17.3	17.1	14.6	11.8	7.6	5.4
	DBStd	5.29	2.90	3.23	2.85	2.83	2.84	2.52	2.32	2.31	2.38	2.72	3.15	3.73
	HDD10.0	707	168	140	98	46	14	1	0	0	0	12	82	147
	HDD18.3	2870	426	371	349	271	200	111	48	53	114	205	321	401
	CDD10.0	916	1	3	7	24	72	143	226	218	139	67	12	4
	CDD18.3	38	0	0	0	0	1	3	16	14	3	1	0	0
	CDH23.3	221	0	0	0	2	7	28	107	55	17	5	0	0
	CDH26.7	26	0	0	0	0	1	4	14	6	1	1	0	0
Wind	WSAvg	5.0	5.5	5.6	5.3	4.9	4.8	4.3	4.2	4.8	4.8	5.3	5.3	5.8
Precipitation	PrecAvg	651	50	45	42	47	46	62	58	53	64	60	64	60
	PrecMax	854	90	76	92	138	95	234	113	116	128	113	127	106
	PrecMin	405	10	7	10	9	9	11	24	6	18	14	42	31
	PrecStd	112	22	18	21	32	24	51	29	29	31	24	26	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.1	17.8	21.1	24.1	27.2	28.8	27.2	25.2	22.1	16.9	14.2
		MCWB	11.6	11.7	12.1	14.4	18.1	20.3	20.3	19.3	18.8	17.4	15.1	12.4
	2%	DB	11.9	12.9	15.1	18.1	21.2	23.2	26.2	25.0	22.9	19.0	14.9	13.1
		MCWB	10.2	10.6	11.3	12.9	15.8	17.3	19.1	18.5	17.2	16.2	13.2	11.5
	5%	DB	10.2	11.2	13.2	16.2	19.2	21.8	24.1	23.1	20.9	17.1	13.2	12.0
		MCWB	8.9	9.2	10.2	12.1	14.3	16.5	17.9	17.1	16.6	14.9	11.8	10.7
	10%	DB	9.0	10.0	11.9	14.8	17.8	20.1	22.9	22.0	19.0	16.0	12.1	10.9
		MCWB	8.1	8.6	9.1	11.2	13.3	15.8	17.2	16.6	15.3	13.8	10.7	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.1	12.9	15.5	18.3	20.9	21.2	20.2	19.6	17.7	15.1	12.9
		MCDB	13.8	14.0	15.7	20.2	23.3	26.0	27.0	24.5	23.4	21.5	16.5	14.0
	2%	WB	10.5	10.9	11.7	13.6	16.6	18.6	19.8	19.3	18.1	16.5	13.5	11.8
		MCDB	11.6	12.6	14.6	17.0	20.6	22.2	24.3	23.5	21.6	18.7	14.8	13.1
	5%	WB	9.2	9.8	10.6	12.7	15.2	17.3	18.8	18.3	16.9	15.3	12.3	10.9
		MCDB	9.9	11.1	13.0	15.9	18.5	20.6	22.8	22.3	19.7	17.1	13.5	12.1
	10%	WB	7.9	8.6	9.6	11.4	13.9	16.1	17.9	17.5	15.9	14.1	11.2	9.8
		MCDB	8.7	9.8	11.6	14.3	16.7	19.3	21.5	20.8	18.4	15.7	12.3	10.6
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.7	7.4	8.3	8.4	8.3	9.0	8.7	8.1	6.5	5.6	5.1
		MCDBR	6.3	6.6	9.5	11.3	11.5	11.2	11.7	10.3	10.3	7.5	5.9	5.7
		MCWBR	5.6	4.9	6.4	7.1	6.7	6.5	5.9	4.9	5.8	5.2	5.1	5.1
	5% WB	MCDBR	6.2	6.5	9.4	10.8	10.5	10.0	10.5	8.8	8.7	7.0	5.9	5.7
		MCWBR	5.8	5.1	6.7	7.1	6.8	6.5	6.1	4.8	5.5	5.1	5.2	5.2
Clear-Sky Solar Irradiance	taub	0.324	0.341	0.372	0.398	0.397	0.404	0.406	0.394	0.372	0.353	0.326	0.311	
	taud	2.350	2.388	2.301	2.244	2.294	2.297	2.304	2.358	2.420	2.448	2.403	2.325	
	Ebn at Noon	638	756	803	829	851	846	837	826	799	733	634	574	
	Edh at Noon	58	76	102	122	122	123	120	108	90	71	55	50	
All-Sky Solar Radiation	RadAvg	0.66	1.33	2.44	3.79	4.77	4.92	4.79	3.97	2.90	1.65	0.81	0.52	
	RadStd	0.07	0.16	0.28	0.44	0.32	0.44	0.48	0.26	0.26	0.14	0.09	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page